4 Pollination

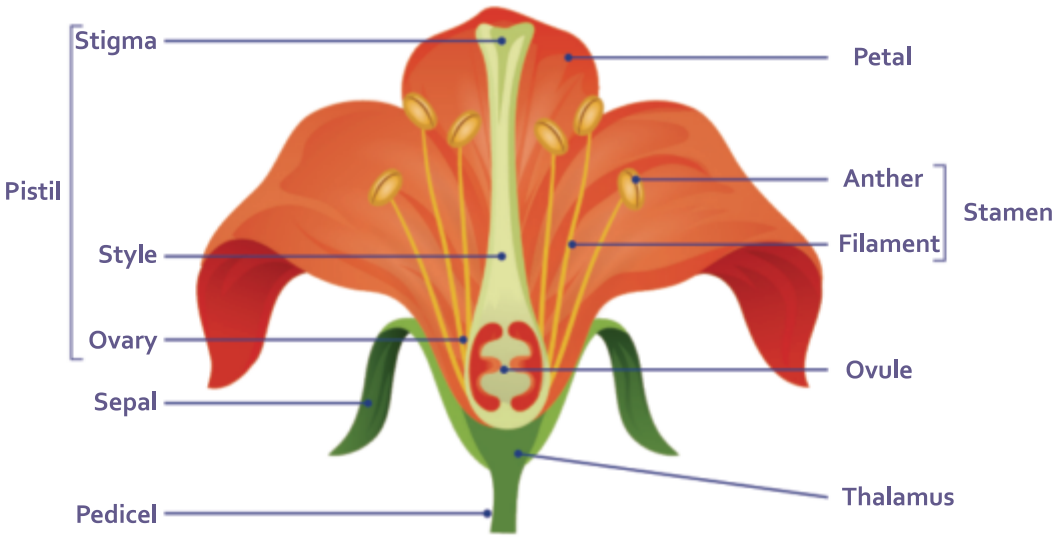
Most plants contain colourful flowers. These flowers attract animals and insects towards the plant. A flower is the reproductive part of the plant. It produces fruits and seeds. Thus, flowers cause new plants to grow. We will learn about the importance of the flower and the process of pollination in this chapter.

Learn about

- Parts of a flower
- Bisexual and monosexual flowers
- Pollination
- Process of pollination
- Self and cross-pollination

Parts of a Flower

A flower generally contains four parts: **calyx**, **corolla**, **androecium**, and **gynoecium**. These are also called the four whorls of a flower. *The stalk of a flower is called the **pedicel**. The swollen end of the pedicel is called the **thalamus**.* Let us study about the different parts of a flower.



Parts of a flower

Calyx

*The outermost whorl of a flower is called the **calyx**.* It contains a set of green leaf-like structures called **sepals**. There may be five or more sepals in a flower.

Function The main function of sepals is to protect the flower in the bud stage. They can also prepare food for the plant.

Corolla

*The second whorl of a flower is called the **corolla**.* It generally consists of a ring of brightly coloured leaf-like structures called **petals**.

Function Since petals are generally brightly coloured, they attract insects that help in pollination. They may also produce different scents to attract insects towards the flower.

Androecium

*The third whorl of a flower is the **androecium**.* It is the collection of **stamens**. *Each stamen has a long slender stalk called **filament**.* An **anther** is present at the tip of the stamen. Anthers produce dust-like particles called **pollen grains**. Pollen grains contain the male gametes or male reproductive cells.

Function The stamen is the male reproductive unit of a flower. Stamens of the flower are collectively called androecium.

Stamen—Male reproductive part of a plant

Gynoecium

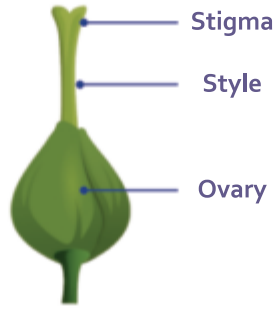
The **gynoecium** is usually located at the centre of a flower. It is a collection of the female reproductive units called carpels or **pistils**. A gynoecium may consist of a single pistil or a group of pistils. A pistil has three parts—stigma, style, and ovary.

***Stigma** is the sticky, swollen tip of the pistil.* It is responsible for receiving the pollen grains.

***Style** is a thin, long, tube-like structure that connects the stigma to the ovary.*

***Ovary** is the swollen base of the pistil.* It contains ovules. The ovules contain eggs or the female gametes.

Function After fertilization, the ovary forms the fruit of the plant, whereas the ovules form the seeds.



Carpel—Female reproductive part of a plant

▶ Bisexual and Monosexual Flowers

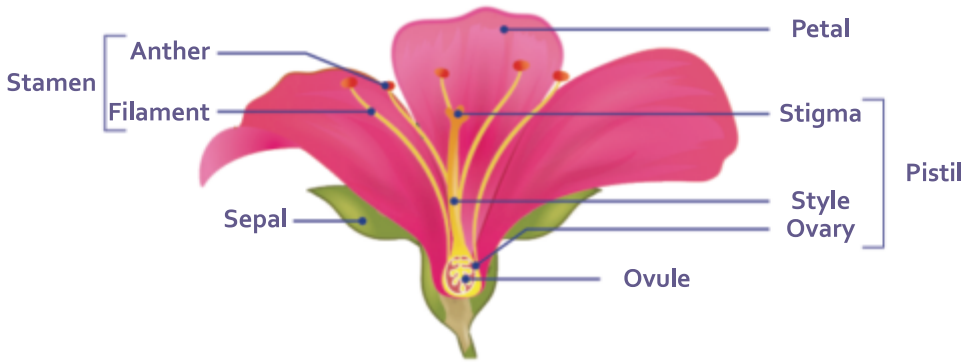
All flowers may or may not contain both male and female reproductive parts. On the basis of the presence of one or both reproductive parts, flowers may be classified as monosexual (or unisexual) and bisexual.

Bisexual Flowers

These flowers contain both the male and female reproductive parts (stamen and pistil) of the plant. They are also called complete flowers. *Tulip*, rose, and *Hibiscus* are some examples of bisexual flowers.

Fact File

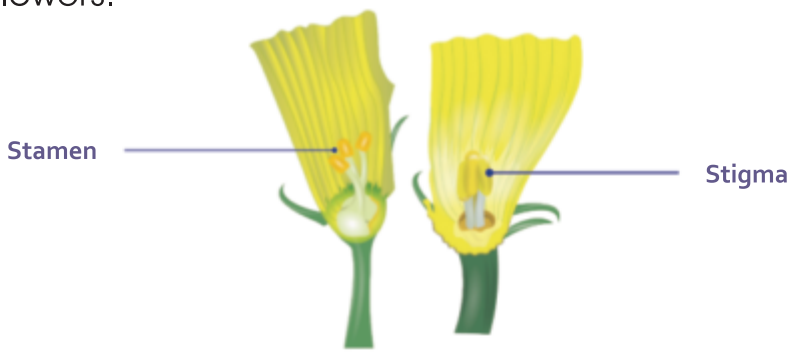
The moonflowers bloom only at night, and close during the day.



Longitudinal section of a bisexual flower

Monosexual Flowers

These flowers contain either the male reproductive part (stamen) or the female reproductive part (pistil) of the plant. They are also called incomplete flowers. Flowers of watermelon, pumpkin, cucumber, and papaya are some examples of monosexual flowers.



Longitudinal section of a unisexual flower showing male and female reproductive parts

Activity

Aim: To study the reproductive parts of a *Hibiscus* flower—stamen and pistil

Materials required: A *Hibiscus* flower (preferably in bud condition), forceps, a dissecting microscope, a needle with handle, a soft hair brush, and a slide

A dissecting microscope

Longitudinal section of a flower

Procedure:

1. Observe the flower closely. Using the forceps, carefully remove two or three sepals (take help from your teacher to learn how to use forceps correctly).
2. Next, carefully remove any two adjacent large red petals of the flower.
3. **To study the stamen:** The stamens are located just inside the petals. A *Hibiscus* flower has many stamens which are united to form a tube-like structure called the staminal tube. Using the forceps, carefully pluck one or two stamens from the top portion of this tube.
4. On the slide, place a stamen in a drop of water (to prevent drying) with the help of a brush.
5. Put it under the magnifying lens of a dissecting microscope to observe it.
6. **To study the pistil:** With the help of your teacher, use the forceps and needle to split open the staminal tube.

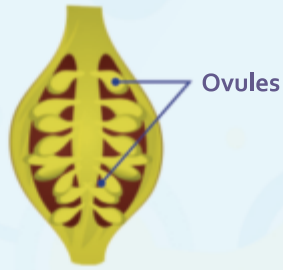
Observations:

1. The stamen has two parts—a small kidney-shaped terminal part is attached to a thin thread-like structure. When we split open the kidney-shaped part using forceps and a needle, we see many yellow tiny particles inside it.
2. Inside the staminal tube, there is a long, slender red tube called the style, which is swollen at the base and divides into five small bulbs at the top. This is the female reproductive part known as the pistil. Ask your teacher to slice the ovary lengthwise to see the ovules inside.

Conclusion: The male reproductive part of a flower (stamen) has two parts—the terminal kidney-shaped structure called the anther attached to a thread-like structure called a filament.

The female reproductive part of a flower (pistil) has the following three parts:

- The lowermost swollen round portion is called the ovary. The ovary contains bead-like structures called ovules.
- The long, slender, tube-like structure in the middle of the pistil is called the style.
- The top portion of the style is further divided into five small bulbs called the stigma.



Ovary with visible ovules

REMEMBERING

Questions

Write *T* for True and *F* for False.

1. The outermost whorl of a flower is called the corolla. .....
2. The androecium is called the male reproductive part of a plant. .....
3. A pistil has three parts—stigma, style, and ovary. .....
4. *Tulip* is an example of a monosexual flower. .....

▶ Pollination

The transfer of pollen grains from the anther to the stigma by various agents is called ***pollination****. The various forces or organisms that move pollen grains within a plant or from one plant to another are called* ***agents of pollination****.* Wind, water, birds, and insects are examples of agents of pollination. Pollination leads to the formation of seeds and fruits in a plant, from which new generations of plants can grow.

21st CENTURY SKILLS SDG

Case Study

Bees help in pollinating many types of plants. When bees collect nectar from flowers, they transfer pollen from one flower to another. This process, called pollination, helps plants produce fruits and seeds. Recently, scientists have observed a decline in bee populations due to pesticide use, habitat loss, and pollution. To help bees, we should plant more flowers, reduce pesticide use, and create safe habitats for bees.

Based on the above information, answer the following questions:

a. What are some of the reasons for the decline in bee populations?

b. What actions can we take to help protect and support bee populations?

Process of Pollination

Pollination is the first step in the process of seed formation. The male reproductive cells (or gametes) are present in the pollen grains that are inside the anther. The female reproductive cells (or gametes) are present in the ovules contained in the ovary. Both the male and female gametes must come in contact with each other for reproduction to take place. When the anthers mature, they split open and the pollen grains are set free. The pollen grains may fall on the stigma of the same flower (in case of bisexual flowers). They may also be carried to the stigma of another flower belonging to the same kind by the different agents of pollination.

Think and Discuss

If pollination never occurred, how would it affect plants, animals, and humans?

▶ Types of Pollination

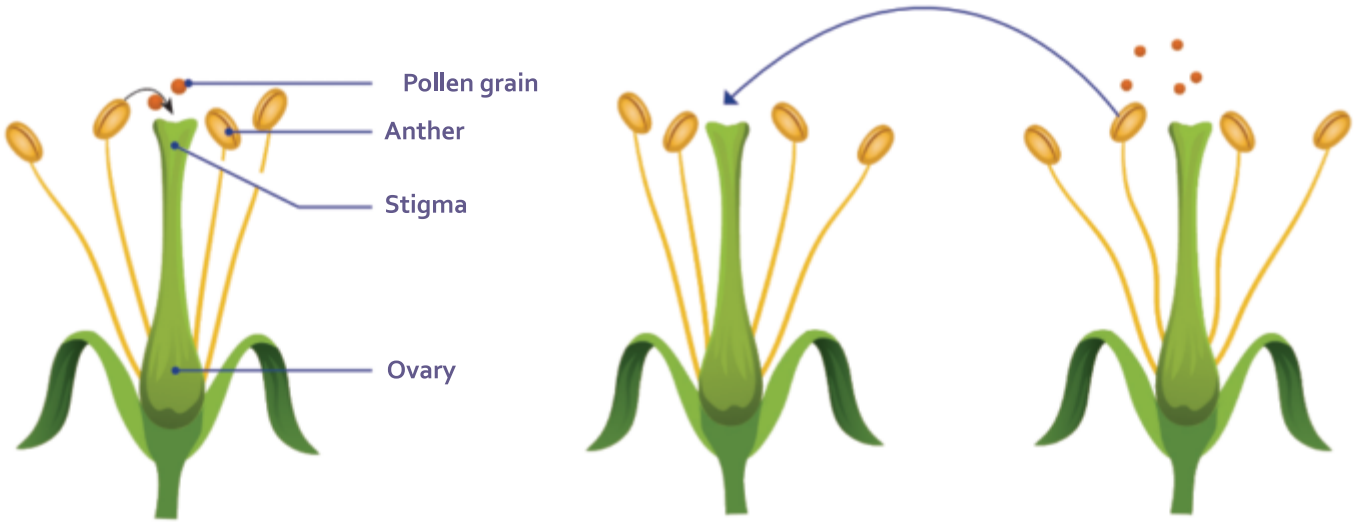
Pollination is of the following two types.

Self-pollination

When the pollen grains from the anther of one flower fall on the stigma of the same flower, or of another flower present on the same plant, the process is called ***self-pollination****.* Self-pollination is commonly seen in pea plants.

Cross-pollination

When the pollen grains from the anther of one flower are transferred to the stigma of another flower, present on a different plant of the same kind, the process is called ***cross-pollination****.* Cross-pollination occurs in most flowering plants.



Self-pollination

Cross-pollination

Wrap Up

- A flower generally consists of four parts: calyx, corolla, androecium, and gynoecium.
- Calyx is the outermost whorl of a flower. It contains a collection of sepals.
- Flowers that bear both the male and female reproductive parts (stamen and the pistil) are called bisexual flowers.
- Flowers that bear only one reproductive part (either the stamen or the pistil) are called monosexual or unisexual flowers.
- The transfer of pollen grains from the anther to the stigma by various agents is called pollination.
- The transfer of pollen grains from the anther to the stigma of the same flower, or of another flower present on the same plant is called self-pollination.
- The transfer of pollen grains from the anther of a flower of one plant to the stigma of a flower of another plant of the same kind is called cross-pollination.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A **Choose the correct option.**

1. Which of the following is not the part of a flower?
   a. Corolla b. Calyx c. Stamen d. Petiole
2. Which of the following is present inside the anther of a stamen?
   a. Pollen grains b. Male reproductive cells
   c. Male gametes d. All are correct
3. What does the stigma of the flower receive during pollination?
   a. Style b. Pollen grains c. Ovules d. Seeds
4. Which of the following is not an example of a monosexual flower?
   a. Hibiscus flower b. Papaya flower
   c. Pumpkin flower d. Watermelon flower

5. Pollination leads to the formation of ..... .
   a. Ovules b. Seeds c. Male gametes d. Pollen grains

B Assertion and Reasoning questions

1. **Assertion (A):** The stamen produces pollen grains containing male gametes.
   Reason (R): The stamen is the male reproductive unit of a flower.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** Petals are brightly coloured in flowers to attract insects.
   Reason (R): Brightly coloured petals serve to attract insects that aid in pollination.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Name the following.

1. The colourful and reproductive part of a plant .....
2. The various parts of a flower .....
3. The whorl that protects the flower in its bud stage .....
4. The long and slender part of the stamen .....
5. The female reproductive unit of the gynoecium .....

SECTION II

D Short answer questions

1. What are the two reproductive parts of a flower?
2. What is the importance of corolla present in a flower?
3. Mention the various parts of the androecium.
4. What is a bisexual flower?
5. Define unisexual flowers with examples.

E Long answer questions

1. Describe the different parts of a flower.
2. Define pollination. Also, mention the two types of pollination.

Picture Study

Look at the picture of the flower below and answer the following questions.

1. Label the numbered parts.
2. In which part are ovules present?

   .....

3. What is the function of part (iii)?

   .....

4. What does part (iv) produce?

   .....

5. Name the process that is involved in the transfer of pollen from part (iv) to part (vi).

   .....

My Learning Corner

A Think about

Renu asks her teacher, "Sir, can you please explain why the stigma of gynoecium is sticky in most flowers?" Can you explain why?

.....

.....

B Try out

1. Click pictures of flowers that are growing in your home, neighbourhood, and adjoining parks. Paste them in your Science scrapbook. Find out their names from gardeners, elders, and Internet. Also, try to find whether they are bisexual or monosexual flowers. Draw and complete the table as given below.

S. no	Picture of flower	Name of flower	Bisexual/Monosexual flower
a.		.....	.....
b.		.....	.....

2. Paste pictures of flowers that undergo self-pollination and cross-pollination (five each), in your Science scrapbook.
3. Draw diagram of a flower showing various parts. Also, label them.

ART INTEGRATION

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• identify various parts of a flower and label each part.	
• draw diagrams of each part of a flower (after observation).	
• locate parts of a flower involved in the process of pollination.	
• explain/discuss the process of pollination by using technical terms.	
• differentiate between self- and cross-pollination and provide examples of each type (showing pictures).	

Worksheet

Different parts of a flower are given below. Draw them individually, in the space provided. Also, write their main function.

S. no	Part of the flower	Draw diagram of the part	Function of the part
1.	Sepal		....
2.	Petal		....
3.	Stamen		....
4.	Pistil		....